# Companion to Bird & de Moor, Algebra of Programming

# §10. Greedy Algorithms

## §10.1. Theory

(10.1a)

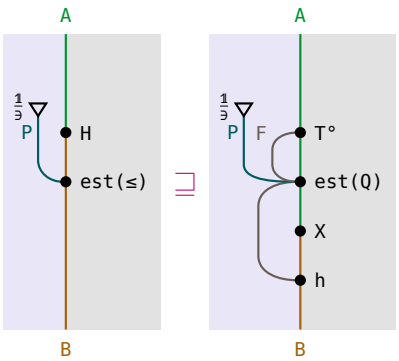
h, T, ≤, H, M as in M ≜ $\frac{H}{\ni}$ est(≤) ; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\ni}$ returns, so that $\frac{T^\circ}{\ni}$ est(Q) is entire.

**Map(h); h Pres ≤; ≤² ⊑ ≤; ThinCondition(T,h,≤,Q)** Theorem 10.1

**⇒ (μX : $\frac{T^\circ}{\ni}$ est(Q) F(X) h) ⊑ M**

the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

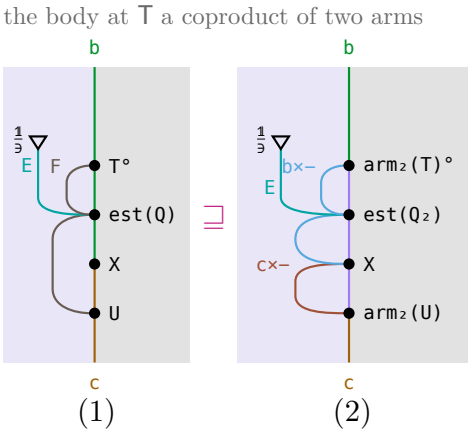
H≜(T)°(h) — ≜ H, Q




(1) (2)

| | | |
|---|---|---|
| 1 | M | |
| 2 | ⊒ $\frac{T^\circ}{\ni}$ est(Q) F(X) h | |

the body at T a coproduct of two arms

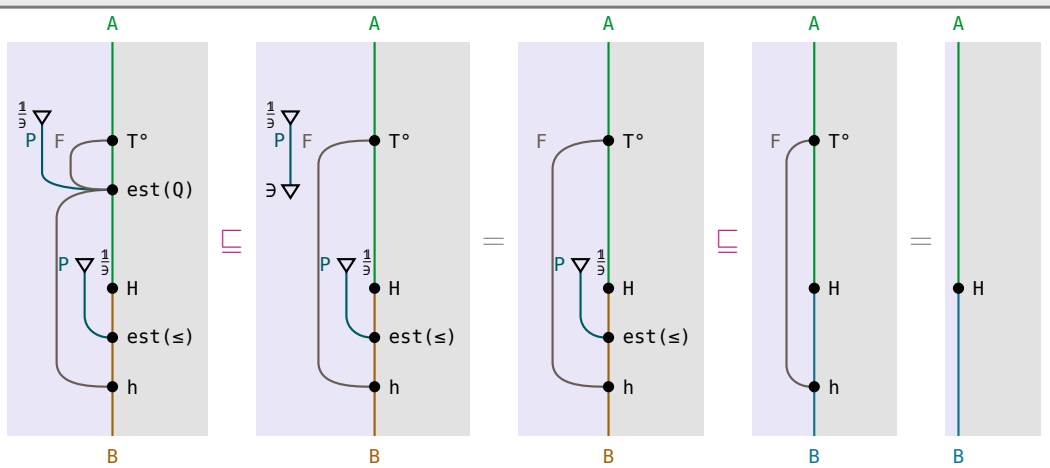



(1) (2)

| | | |
|---|---|---|
| 1 | $\frac{T^\circ}{\ni}$ est(Q) F(X) U | Proposition 10.1, V₂V₁°=0 |
| 2 | ⊒ $\frac{arm_2(T)^\circ}{\ni}$ est(Q₂) (X × 𝟙) arm₂(U) | |

**$\frac{T^\circ}{\ni}$ est(Q) F(M) h ⊑ H** (10.1b)

taking the input apart in one Q-extreme way, solving the parts by M and assembling by h returns only what H returns



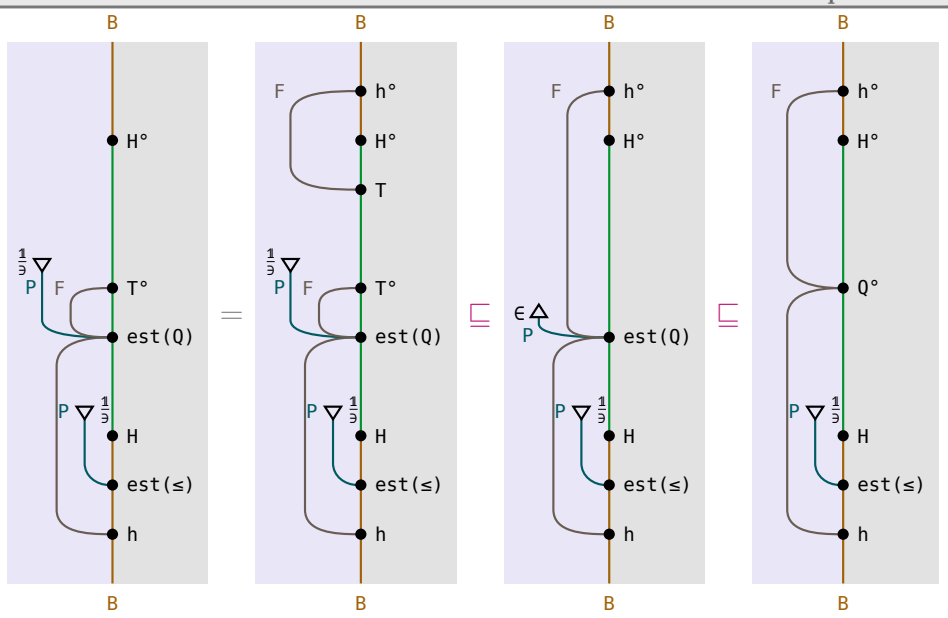

(1) (2) (3) (4) (5)

| | | |
|---|---|---|
| 1 | $\frac{T^\circ}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(≤)) h | est(Q) ⊑ ∋ |
| 2 | ⊒ $\frac{T^\circ}{\ni}$ ∋ F($\frac{H}{\ni}$ est(≤)) h | $\frac{R}{\ni}$ ∋ = R |
| 3 | = T° F($\frac{H}{\ni}$ est(≤)) h | $\frac{H}{\ni}$ est(≤) ⊑ H |
| 4 | ⊑ T° F(H) h | T° F(H) h = H |
| 5 | = H | |

**Map(h); h Pres ≤; ≤² ⊑ ≤; ThinCondition(T,h,≤,Q)** (10.1c)

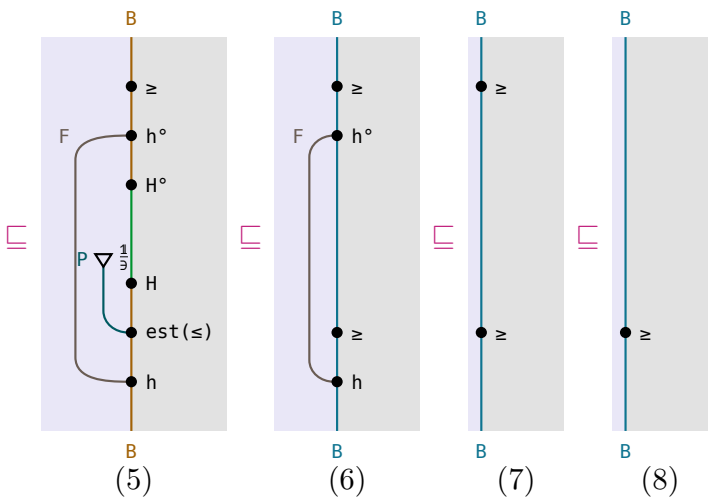
**⇒ H° $\frac{T°}{∋}$ est(Q) F(M) h ⊑ ≥**

**H°** followed by the greedy body at **M** is **⊑≥**: an answer of the body is never worse than an answer of **H** to the same input




(1) (2) (3) (4)

| | | |
|---|---|---|
| 1 | H° $\frac{T°}{∋}$ est(Q) F($\frac{H}{∋}$ est(≤)) h | H° = h° F(H°) T |
| 2 | = h° F(H°) T $\frac{T°}{∋}$ est(Q) F($\frac{H}{∋}$ est(≤)) h | T $\frac{T°}{∋}$ ⊑ ∈ |
| 3 | ⊑ h° F(H°) ∈ est(Q) F($\frac{H}{∋}$ est(≤)) h | ∈ est(Q) ⊑ Q° |
| 4 | ⊑ h° F(H°) Q° F($\frac{H}{∋}$ est(≤)) h | ThinCondition(T,h,≤,Q)<br>⇒ h° F(H°) Q° ⊑ ≥ h° F(H°) |



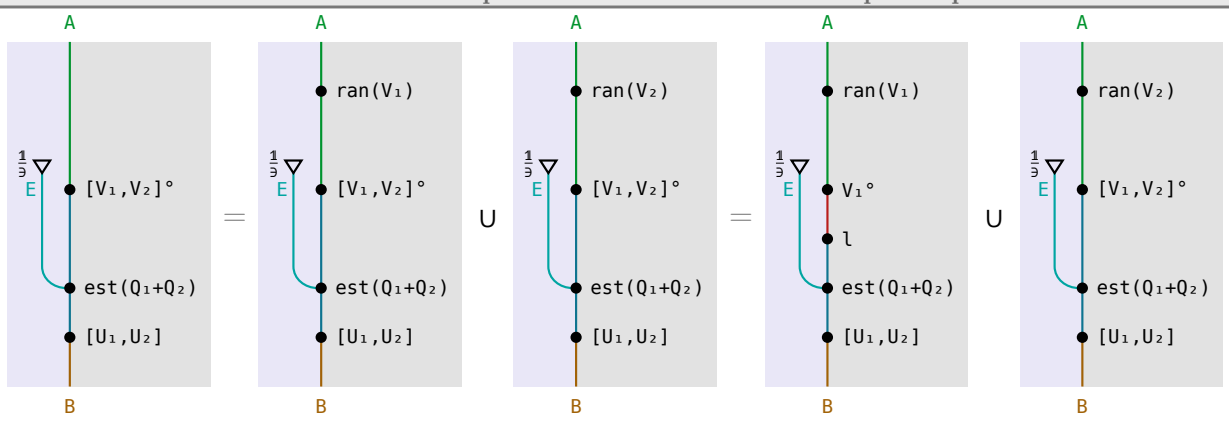

(5) (6) (7) (8)

| | | |
|---|---|---|
| 5 | ⊑ ≥ h° F(H° $\frac{H}{∋}$ est(≤)) h | H° $\frac{H}{∋}$ est(≤) ⊑ ≥ |
| 6 | ⊑ ≥ h° F(≥) h | Map(h); h Pres ≤<br>⇒ h° F(≥) h ⊑ ≥ |
| 7 | ⊑ ≥² | ≤² ⊑ ≤<br>⇒ ≥² ⊑ ≥ |
| 8 | ⊑ ≥ | |

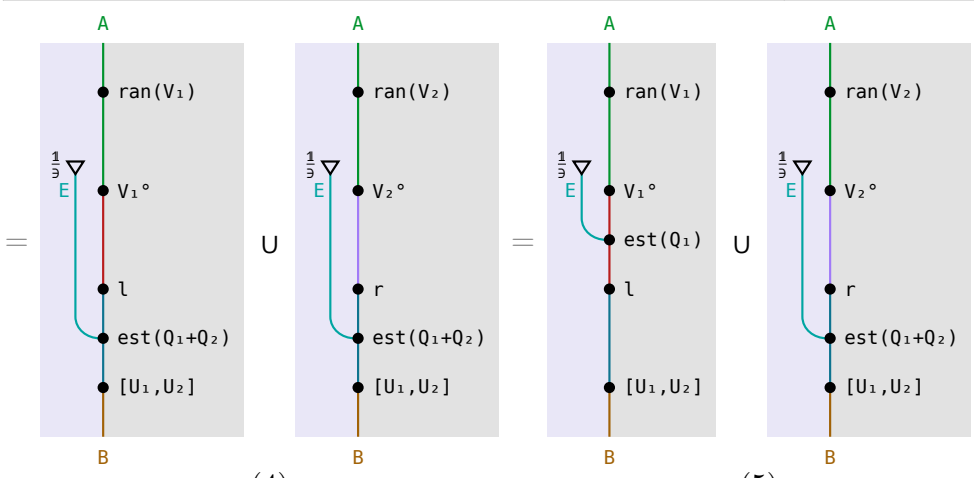
**Proposition 10.1**

$$V_2\ V_1{}^\circ = \emptyset \Rightarrow \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2] = (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ U_1) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ U_2)$$

when $V_1$ and $V_2$ have disjoint ranges, the greedy step over $[V_1,V_2]$ runs the $V_1$ step on inputs $V_1$ reaches and the $V_2$ step on inputs $V_2$ reaches




| | | |
|---|---|---|
| 1 | $\frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2]$ | $\frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2] = (ran(V_1)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2])$ |
| 2 | $= (ran(V_1)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2])$ | $V_2\ V_1{}^\circ = \emptyset \Rightarrow ran(V_1)\ \frac{[V_1,V_2]^\circ}{\ni} = ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ P(l)$ |
| 3 | $= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ P(l)\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2])$ | $V_2\ V_1{}^\circ = \emptyset \Rightarrow ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni} = ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ P(r)$ |




| | | |
|---|---|---|
| 4 | $= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ P(l)\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ P(r)\ est(Q_1 + Q_2)\ [U_1,U_2])$ | $P(l)\ est(Q_1 + Q_2) = est(Q_1)\ l$ |
| 5 | $= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ l\ [U_1,U_2]) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ P(r)\ est(Q_1 + Q_2)\ [U_1,U_2])$ | $P(r)\ est(Q_1 + Q_2) = est(Q_2)\ r$ |
| 6 | $= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ l\ [U_1,U_2]) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ r\ [U_1,U_2])$ | $l\ [U_1,U_2] = U_1$ |
| 7 | $= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ U_1) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ r\ [U_1,U_2])$ | $r\ [U_1,U_2] = U_2$ |
| 8 | $= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ U_1) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ U_2)$ | |

## §10.2. The detab-entab problem

### §10.2.1. Detab

(10.2.1a)

| | definition | type | meaning |
|---|---|---|---|
| 1 | `(\|[nil,expand]\|) = detab` | `String→ String` | replace every tab by blanks up to the next tab stop; `H=detab°` |
| 2 | `expand(xs,a) ≜ if a = tb then fill(xs) else snoc(xs,a)` | `String` | a tab fills the line to the next stop; any other character is appended |
| 3 | `fill(xs) ≜ xs⧺blanks(n−col(xs) % n)` | `String` | append blanks up to the next multiple of `n` columns |
| 4 | `col(wrap(a)) = 0`<br>`col(snoc(x,a)) = if a = nl then 0 else col(x) + 1` | `String→ℕ` | the column the string ends at: characters since the last newline |
| 5 | `xs R ys ⇔ length(xs) ≤ length(ys)` | `[Char]→ [Char]` | `x` is no longer than `y` |
| 6 | `a U b ⇔ a = tb ∨ a = b` | `Char→Char` | `a` is the tab or `a=b` |
| 7 | `xs V ys ⇔ prefix(xs,ys) ∧ fill(xs) = fill(ys)` | `[Char]→ [Char]` | a prefix that fills to the same string |
| 8 | `Q ≜ 𝟙 + (V × U)` | `F[Char]→ F[Char]` | compare two decompositions by `V` on the strings and `U` on the characters |
| 9 | `unfill` | `String→ String` | the shortest prefix that fills to the same string |
| 10 | `tbc` | `String→ℕ` | the number of trailing blanks |
| 11 | `triple ≜ ⟨unfill entab,⟨tbc,col⟩⟩` | `[Char]→ String×(ℕ×ℕ)` | what `entab` keeps: the unfilled output, its trailing blank count and its column |

```
unfill(wrap(a)) = []
unfill(snoc(x,a)) = if a = blank ∧ col(snoc(x,a)) % n ≠ 0 then unfill(x) else snoc(x,a)
```

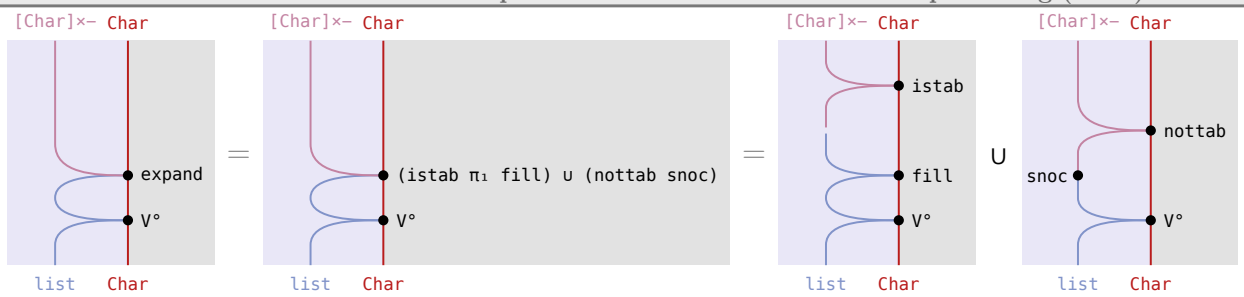
```
tbc(wrap(a)) = 0
tbc(snoc(x,a)) = if a = blank ∧ col(snoc(x,a)) % n ≠ 0 then tbc(x) + 1 else 0
```

## §10.2.2. Derivation

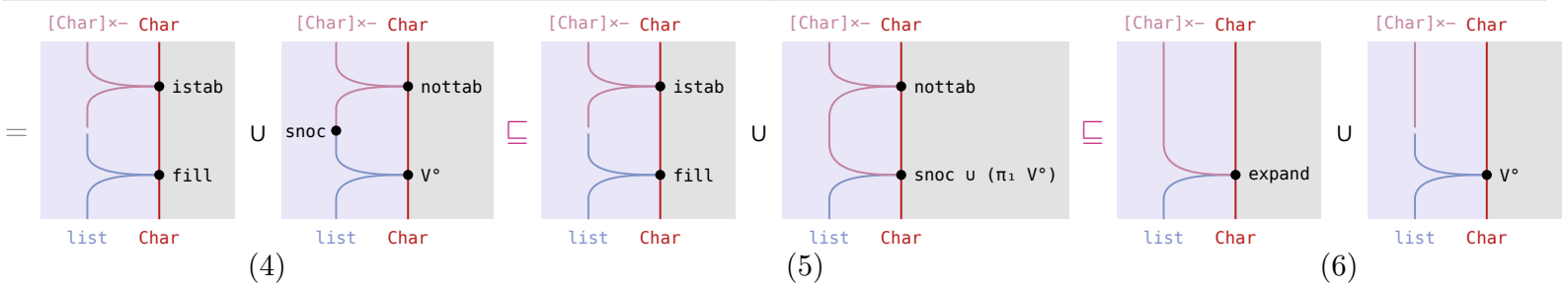
**0 < n** (10.2.2a)

**⇒ expand V° ⊑ expand ∪ (π₁ V°)**

shortening the output of one **expand** step to a **V**-smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string (**π₁V°**)




| | | |
|---|---|---|
| 1 | expand V° | expand = (istab π₁ fill) ∪ (nottab snoc) |
| 2 | = ((istab π₁ fill) ∪ (nottab snoc)) V° | (S ∪ T) R = (S R) ∪ (T R) |
| 3 | = (istab π₁ fill V°) ∪ (nottab snoc V°) | 0 < n; blank ≠ nl<br>⇒ fill V° = fill |

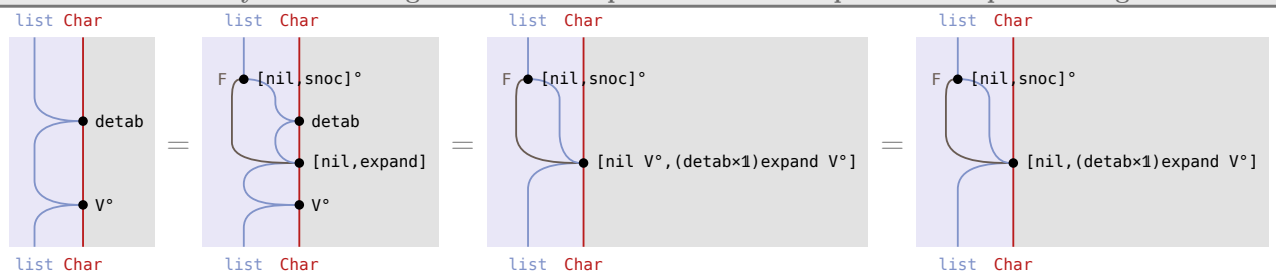



| | | |
|---|---|---|
| 4 | = (istab π₁ fill) ∪ (nottab snoc V°) | 0 < n; blank ≠ nl<br>⇒ snoc V° ⊑ snoc ∪ (π₁ V°) |
| 5 | ⊑ (istab π₁ fill) ∪ (nottab (snoc ∪ (π₁ V°))) | (istab π₁ fill) ∪ (nottab (snoc ∪ X)) ⊑ expand ∪ X |
| 6 | ⊑ expand ∪ (π₁ V°) | |

**`0 < n`** (10.2.2b)

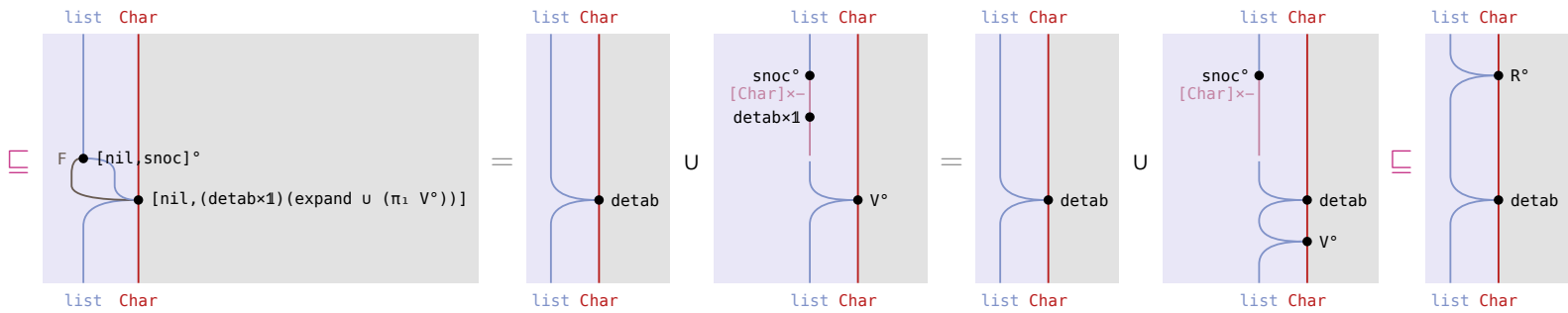
**`⇒ detab V° ⊑ R° detab`**

any **V**-shortening of **detab**'s output is **detab**'s output on an input no longer than the given one




(1) (2) (3) (4)

| | | |
|---|---|---|
| 1 | `detab V°` | `detab = [nil,snoc]° F(detab) [nil,expand]` |
| 2 | `= [nil,snoc]° F(detab) [nil,expand] V°` | `[R,expand] Z = [R Z,expand Z]` |
| 3 | `= [nil,snoc]° F(detab) [nil V°,expand V°]` | `nil V° = nil` |
| 4 | `= [nil,snoc]° F(detab) [nil,expand V°]` | (10.2.2a) `0 < n` `⇒ expand V° ⊑ expand ∪ (π₁ V°)` |




(5) (6) (7) (8)

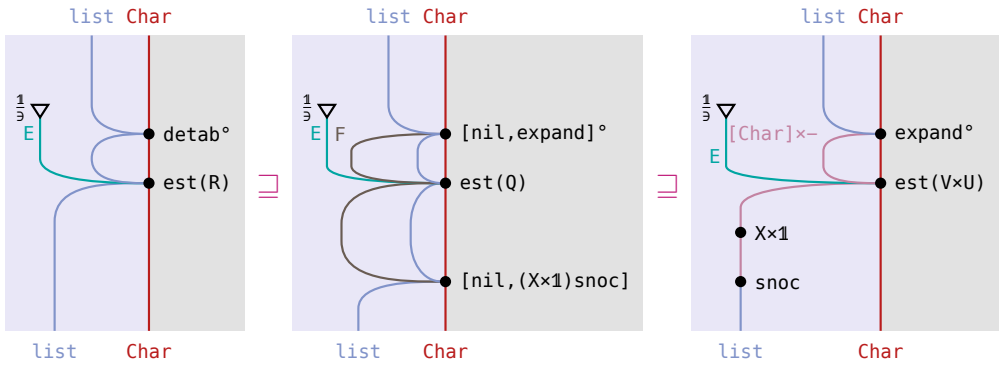
| | | |
|---|---|---|
| 5 | `⊑ [nil,snoc]° F(detab) [nil,expand ∪ (π₁ V°)]` | `[nil,snoc]° F(detab) [nil,expand ∪ X] = detab ∪ (snoc° (detab × 𝟙) X)` |
| 6 | `= detab ∪ (snoc° (detab × 𝟙) π₁ V°)` | `snoc° (detab × 𝟙) π₁ Y = snoc° π₁ detab Y` |
| 7 | `= detab ∪ (snoc° π₁ detab V°)` | `0 < n` `⇒ detab ∪ (snoc° π₁ detab V°) ⊑ R° detab` |
| 8 | `⊑ R° detab` | |

**`0 < n`** (10.2.2c)

**`⇒ (μX : [nil,expand]° ∋ est(Q) F(X) [nil,snoc]) ⊑ detab° ∋ est(R)`**

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

the specification — `detab(wrap(a)) = []`, `detab(snoc(x,a)) = expand(detab(x),a)` , `⦇[nil,expand]⦈ = detab` ; `detab entab=𝟙` and nothing shorter does

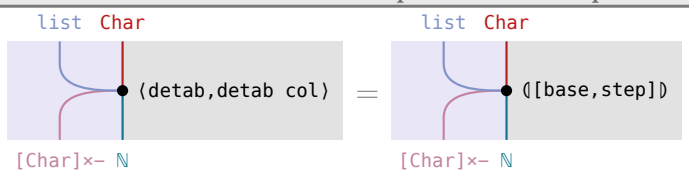



(1) (2) (3)

| | | |
|---|---|---|
| 1 | `detab°/∋ est(R)` | at `Q≜𝟙+(V×U)` : one character of input is decided at each step. `F(⊤,R)α⊑αR`. `detab prefix⊑R° detab` is FALSE , — at `n=8`, `detab [a,b,c,d,e,TB]=[a,b,c,d,e,BL,BL,BL]`, whose prefix `[a,b,c,d,e,BL,BL]` is longer than any input giving it, and `detab V°⊑R° detab` holds. `expand V°⊑expand ∪ (π₁V°)` — shortening the output either leaves the last step alone or discards it |
| 2 | `⊒ [nil,expand]°/∋ est(Q) F(X) [nil,snoc]` | Proposition 10.1: `nil` and `expand` have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop |
| 3 | `⊒ expand°/∋ est(V × U) (X × 𝟙) snoc` | |

**⟨detab,detab col⟩ = ⦇[base,step]⦈** (10.2.2d)

one pass over the input carries the output so far together with its column




(1) (2)

| | | |
|---|---|---|
| 1 | `⟨detab,detab col⟩` | both components are folds over the same input; `step` on `(detab x,col(detab x))` is `(detab,col detab)` one character further — not a tabulated row |
| 2 | `= ⦇[base,step]⦈` | |

**Exercise 10.1**

- `⦇[base,step]⦈(convert(xs),r) ⇔ r = loop(step)(([],0),xs)`
  the fold of `[base,step]` over the input converted to a snoc-list is the left loop of `step` from `base`
- `π₁(loop(step)(([],c₀),xs)) = loop'((f,g),(c₀,xs))`
  Exercise 10.1: when `step` only appends `f(c,a)` to the output and moves the state by `g`, the output of the loop is `loop'(f,g)`, which never carries the output

(10.2.2e)

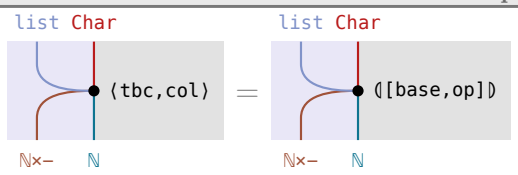
- `a ≠ tb ⇒ expand(ys,b) = snoc(xs,a) ⇔ fill(ys) = snoc(xs,a) ∧ b = tb ∨ (ys,b) = (xs,a)`
  `expand` produces `x#[a]` from a tab after a string that fills to it, or from `x` and `a`
- `0 < n ⇒ (∃ys. fill(ys) = snoc(xs,a)) ⇔ a = blank ∧ col(snoc(xs,a)) % n = 0`
  a string is a `fill` exactly when it ends in a blank on a tab stop

(10.1)

- `entab(xs) = entab(unfill(xs))#blanks(tbc(xs))`
  (10.1): the output of `entab` is that of `unfill x` followed by the trailing blanks held back
- `tbc[] = 0`
  the empty string has no trailing blanks
- `tbc(snoc(xs,a)) = if a = blank ∧ col(snoc(xs,a)) % n ≠ 0 then tbc(xs) + 1 else 0`
  a blank off a tab stop adds one held blank; anything else releases them all

**⟨tbc,col⟩ = ⦇[base,op]⦈** (10.2.2f)

one pass counts the held blanks and the column together

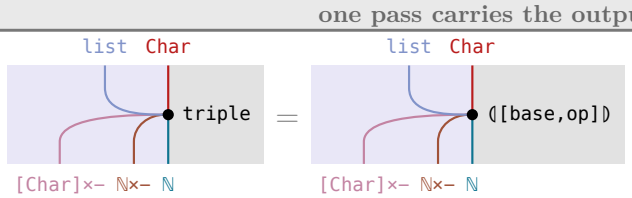



(1) (2)

| | | |
|---|---|---|
| 1 | `⟨tbc,col⟩` | `op` on `(tbc x,col x)` is `(tbc,col)` one character further, from the equations for `tbc` — (10.1) `entab(xs) = entab(unfill(xs))#blanks(tbc(xs))`, `tbc[] = 0`, `tbc(snoc(xs,a)) = if a = blank ∧ col(snoc(xs,a)) % n ≠ 0 then tbc(xs) + 1 else 0` |
| 2 | `= ⦇[base,op]⦈` | |

**triple = ⦇[base,op]⦈** (10.2.2g)

one pass carries the output up to the held blanks, their count and the column

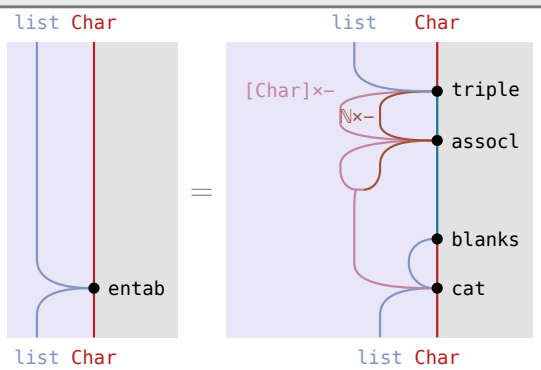



(1) (2)

| | | |
|---|---|---|
| 1 | `triple` | `op` on `triple x` is `triple` one character further: (10.1) where the blanks are released — (10.1) `entab(xs) = entab(unfill(xs))#blanks(tbc(xs))`, `tbc[] = 0`, `tbc(snoc(xs,a)) = if a = blank ∧ col(snoc(xs,a)) % n ≠ 0 then tbc(xs) + 1 else 0` |
| 2 | `= ⦇[base,op]⦈` | |

**entab = triple assocl π₁ (𝟙 × blanks) cat** (10.2.2h)

**entab** is **triple** with the held blanks appended to its output




| | | |
|---|---|---|
| 1 | entab | (10.1) — (10.1) entab(xs) = entab(unfill(xs)) #blanks(tbc(xs)) , tbc[] = 0 , tbc(snoc(xs,a)) = if a = blank ∧ col(snoc(xs,a)) % n ≠ 0 then tbc(xs) + 1 else 0 |
| 2 | = triple assocl π₁ (𝟙 × blanks) cat | |

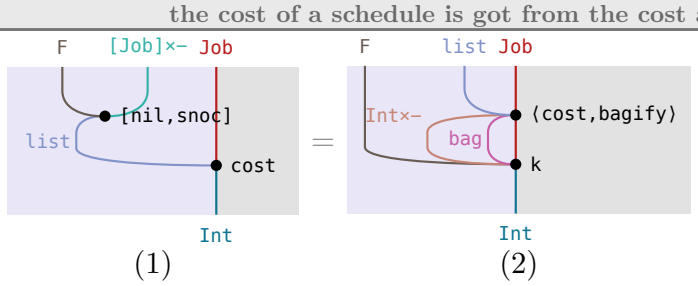
## §10.3. The minimum tardiness problem

(10.3a)

| | definition | type | meaning |
|---|---|---|---|
| 1 | `bag(Job) ≜ Quotient(perm)` | $\mathcal{A}$ | a bag of jobs: a list up to reordering |
| 2 | `snag(p) ≜ π₂(p) :: π₁(p)` | `bag(Job)` | put one job into a bag |
| 3 | `(\|[nil,snag]\|) = bagify` | `[Job]→bag(Job)` | forget the order of a schedule; `H=bagify°` lists every schedule of a bag |
| 4 | `penalty(xs,j) ≜ (sum(ct′)(xs) + ct(j)−dt(j))wt(j)` | `Int` | how late job `j` finishes after the schedule `xs`, times its weight |
| 5 | `cost(wrap(a)) = 0`<br>`cost(snoc(x,a)) = bmax(cost(x),penalty(x,a))` | `[Job]→Int` | the largest penalty of any job in the schedule |
| 6 | `bmax(a,b) ≜ if a ≤ b then b else a` | `Int` | the larger of two numbers |
| 7 | `R ≜ cost ≤ cost°` | `[Job]→[Job]` | `xs` costs no more than `ys` |
| 8 | `p add ys ⇔ add(π₂(p),ys)` | `[Job]×Job→[Job]` | insert `j` anywhere in `xs`; `perm≜bagify` `bagify°=(\|[nil,add]\|)` |
| 9 | `k ≜ [nil,assocr (𝟙×((bagify°×𝟙) penalty)) bmax]` | `F(Int×bag(Job))→ Int` | the cost of a schedule from the cost of its front and the penalty of its last job |
| 10 | `f(l(val)) = 0`<br>`f(r(p)) = (bagify°×𝟙) penalty(p)` | `F(bag(Job))→Int` | the penalty of the last job, read off the bag before it |
| 11 | `u Q v ⇔ f(u) ≤ f(v)` | `F(bag(Job))→ F(bag(Job))` | the last job of `u` gets no larger penalty than that of `v` |
| 12 | `p Q' q ⇔ (bagify°×𝟙) penalty(p) ≤ (bagify°×𝟙) penalty(q)` | `bag(Job)×Job→ bag(Job)×Job` | `Q` on the `snag` summand: compare the penalties of the two last jobs |
| 13 | `g ≜ [zero,penalty]` | `F[Job]→Int` | (10.5): the penalty of the last job, zero for the empty schedule |
| 14 | `m ≜ [zero,π₁ cost]` | `F[Job]→Int` | (10.6), B&dM's `h` renamed as in `(∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j))` ⇒ `(μX : [nil,snag]° ∋ est(Q) F(X) [nil,snoc]) ⊑ bagify° ∋ est(R)` : the cost of the schedule before the last job |

**`[nil,snoc] cost = F(⟨cost,bagify⟩) k`** (10.3b)

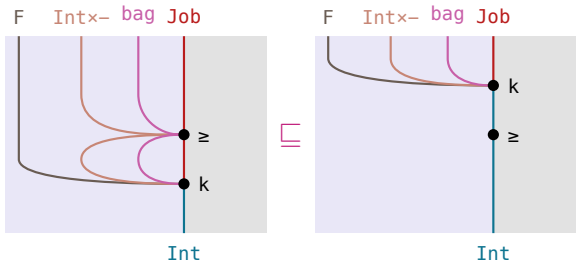
the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job




| | | |
|---|---|---|
| 1 | `[nil,snoc] cost` | `cost (xs#[j])=bmax (cost xs,penalty (xs,j))`, and `penalty(xs,j)` reads only the bag of `xs` — `penalty(xs,j) ≜ (sum(ct′)(xs) + ct(j)−dt(j))wt(j)` |
| 2 | `= F(⟨cost,bagify⟩) k` | |

**`F(≥ × 𝟙) k ⊑ k ≥`** Exercise 10.5

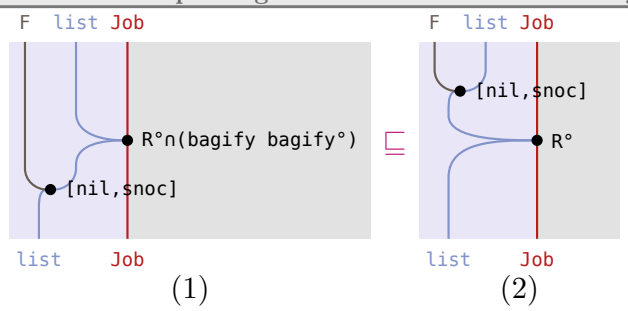
a larger cost before the last job gives a larger cost after it




| | | |
|---|---|---|
| 1 | `F(≥ × 𝟙) k` | `bmax` is monotone in its first argument |
| 2 | `⊑ k ≥` | |

**`F(R° ∩ (bagify bagify°)) [nil,snoc] ⊑ [nil,snoc] R°`** (10.2)

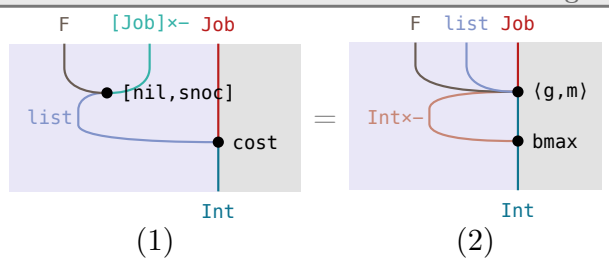
improving the schedule before the last job, among schedules of the same bag, improves the whole schedule




| | | |
|---|---|---|
| 1 | `F(R° ∩ (bagify bagify°)) [nil,snoc]` | Proposition 9.3 at `S≜bagify`, `⩽≜≥` — Proposition 9.3 `Map(cost); ≤ = cost ⩽ cost°; Simple(H°); h cost = F(⟨cost,H°⟩) k; F(⩽ × 𝟙) k ⊑ k ⩽ ⇒ F(≤ ∩ (H° H)) h ⊑ h ≤`, with `[nil,snoc] cost = F(⟨cost,bagify⟩) k` and Exercise 10.5 `F(≥ × 𝟙) k ⊑ k ≥` |
| 2 | `⊑ [nil,snoc] R°` | |

**`[nil,snoc] cost = ⟨g,m⟩ bmax`** (10.4)

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it

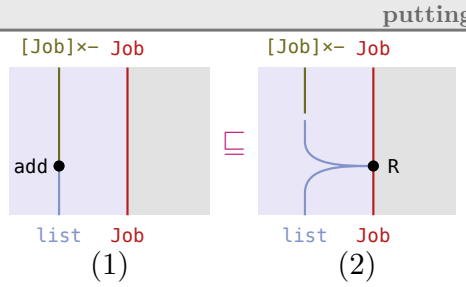



| | | |
|---|---|---|
| 1 | `[nil,snoc] cost` | `cost (xs#[j])=bmax (cost xs,penalty (xs,j))` and `bmax` commutes — `cost(wrap(a)) = 0`, `cost(snoc(x,a)) = bmax(cost(x),penalty(x,a))` |
| 2 | `= ⟨g,m⟩ bmax` | |

**`(∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j))`** (10.7)

**`⇒ add ⊑ π₁ R`**

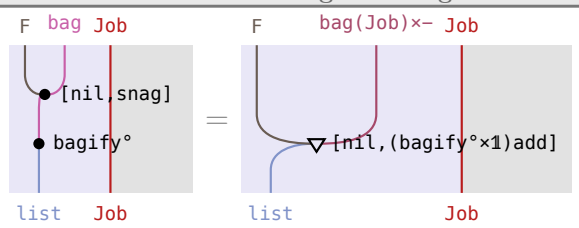
putting one more job anywhere into a schedule never lowers its cost




| | | |
|---|---|---|
| 1 | `add` | induction on where `j` goes: every job after it starts later, so no penalty falls |
| 2 | `⊑ π₁ R` | |

**[nil,snag] bagify° = [nil,(bagify° × 𝟙) add]** (10.8)

the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere

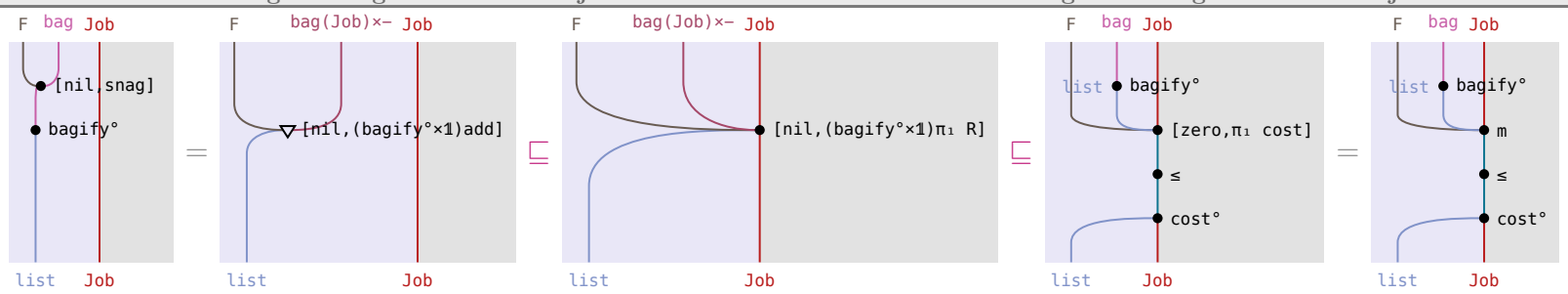



| | | |
|---|---|---|
| 1 | [nil,snag] bagify° | take out the last copy of j and put it back — p add ys ⇔ add(π₂(p),ys) |
| 2 | = [nil,(bagify° × 𝟙) add] | |

**(∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j))** (10.3c)

**⇒ [nil,snag] bagify° ⊑ F(bagify°) m ≤ cost°**

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job

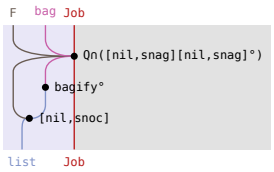



| | | |
|---|---|---|
| 1 | [nil,snag] bagify° | (10.8) [nil,snag] bagify° = [nil,(bagify° × 𝟙) add] |
| 2 | = [nil,(bagify° × 𝟙) add] | (10.7) (∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j)) ⇒ add ⊑ π₁ R |
| 3 | ⊑ [nil,(bagify° × 𝟙) π₁ R] | [nil,π₁ R] ⊑ [zero,π₁ cost] ≤ cost° |
| 4 | ⊑ F(bagify°) [zero,π₁ cost] ≤ cost° | |
| 5 | = F(bagify°) m ≤ cost° | |

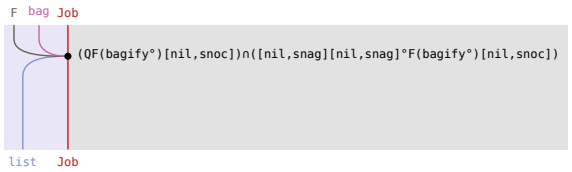
**(∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j))** **(10.3)**

**⇒ (Q ∩ ([nil,snag] [nil,snag]°)) F(bagify°) [nil,snoc] ⊑ F(bagify°) [nil,snoc] R**

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



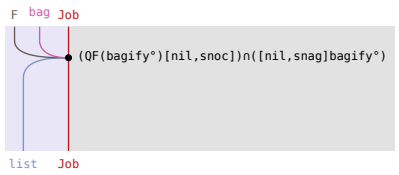

⊑



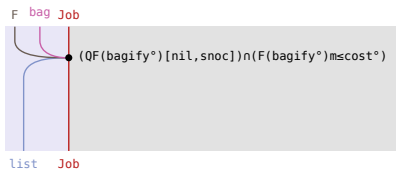

(1) (2)

| | | |
|---|---|---|
| 1 | (Q ∩ ([nil,snag] [nil,snag]°)) F(bagify°) [nil,snoc] | (Q ∩ T) R ⊑ (Q R) ∩ (T R) |
| 2 | ⊑ (Q F(bagify°) [nil,snoc]) ∩ ([nil,snag] [nil,snag]° F(bagify°) [nil,snoc]) | [nil,snag]° F(bagify°) [nil,snoc] = bagify° |

=



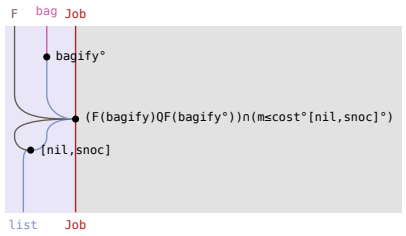

⊑



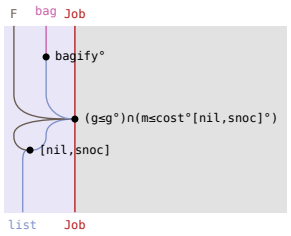

(3) (4)

| | | |
|---|---|---|
| 3 | = (Q F(bagify°) [nil,snoc]) ∩ ([nil,snag] bagify°) | (10.3c) (∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j)) ⇒ [nil,snag] bagify° ⊑ F(bagify°) m ≤ cost° |
| 4 | ⊑ (Q F(bagify°) [nil,snoc]) ∩ (F(bagify°) m ≤ cost°) | (Q F(bagify°) [nil,snoc]) ∩ (F(bagify°) M) ⊑ F(bagify°) ((F(bagify) Q F(bagify°)) ∩ (M [nil,snoc]°)) [nil,snoc] |

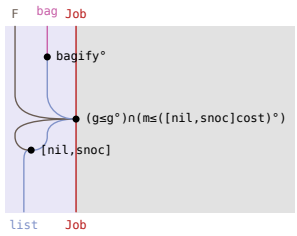
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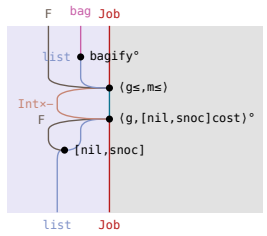

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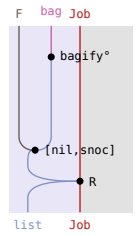

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⊑

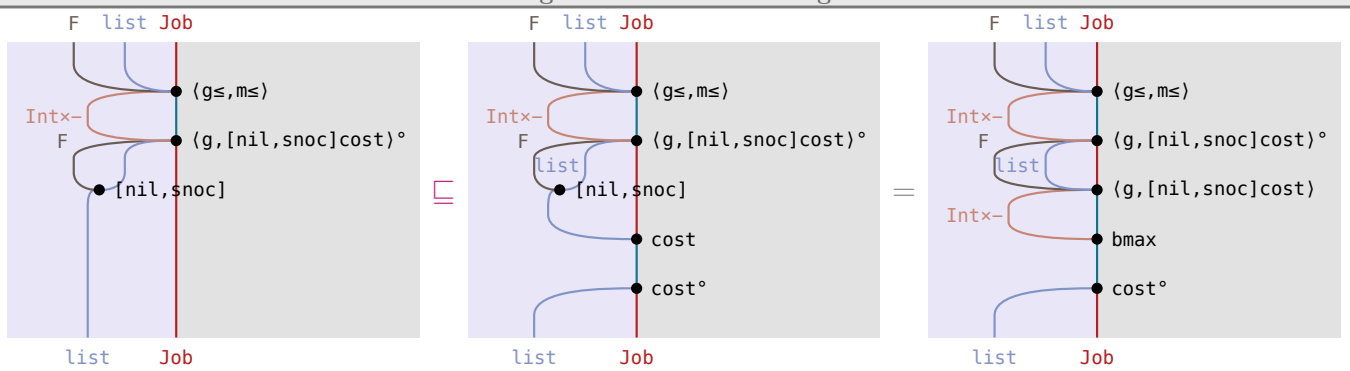



(5) (6) (7) (8) (9)

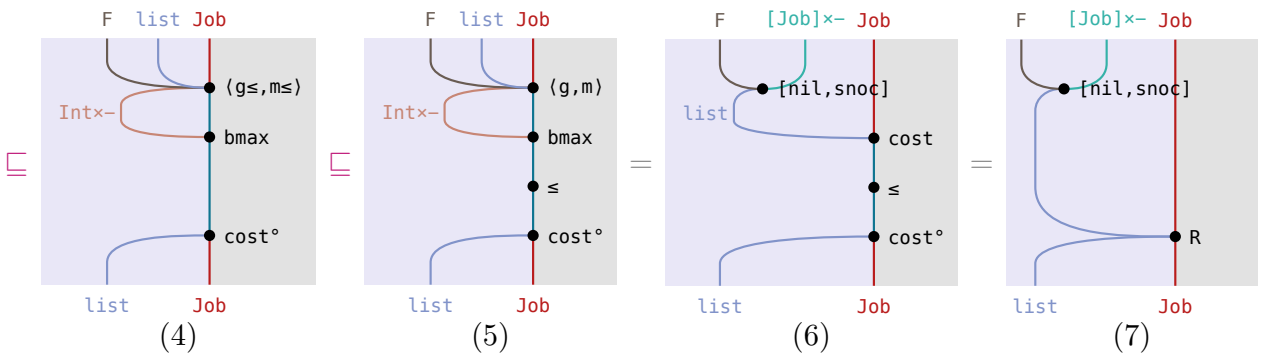
| | | |
|---|---|---|
| 5 | ⊑ F(bagify°) ((F(bagify) Q F(bagify°)) ∩ (m ≤ cost° [nil,snoc]°)) [nil,snoc] | F(bagify) Q F(bagify°) = g ≤ g° |
| 6 | = F(bagify°) ((g ≤ g°) ∩ (m ≤ cost° [nil,snoc]°)) [nil,snoc] | (con cost)° = cost° con° |
| 7 | = F(bagify°) ((g ≤ g°) ∩ (m ≤ ([nil,snoc] cost)°)) [nil,snoc] | ⟨X,Y⟩ ⟨g,S⟩° = ⟨X g°⟩ ∩ ⟨Y S°⟩ |
| 8 | = F(bagify°) ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] | (10.3d) ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] ⊑ [nil,snoc] R |
| 9 | ⊑ F(bagify°) [nil,snoc] R | |

**⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] ⊑ [nil,snoc] R** (10.3d)

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job, gives a schedule costing at most the whole




| | | |
|---|---|---|
| 1 | ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] | 𝟙 ⊑ cost cost° |
| 2 | ⊑ ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] cost cost° | [nil,snoc] cost = ⟨g,[nil,snoc] cost⟩ bmax |
| 3 | = ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° ⟨g,[nil,snoc] cost⟩ bmax cost° | ⟨g,[nil,snoc] cost⟩° ⟨g,[nil,snoc] cost⟩ ⊑ 𝟙 |

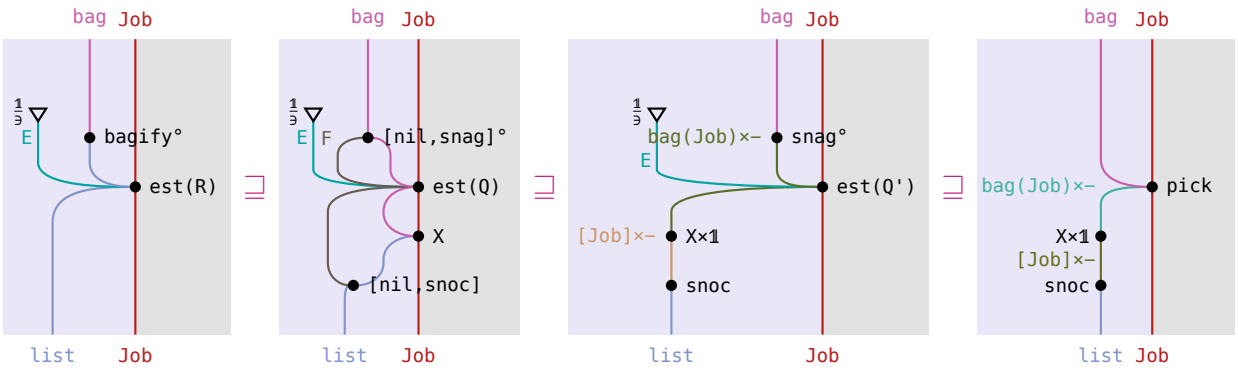



| | | |
|---|---|---|
| 4 | ⊑ ⟨g ≤,m ≤⟩ bmax cost° | ⟨g ≤,m ≤⟩ bmax ⊑ ⟨g,m⟩ bmax ≤ |
| 5 | ⊑ ⟨g,m⟩ bmax ≤ cost° | (10.4) [nil,snoc] cost = ⟨g,m⟩ bmax |
| 6 | = [nil,snoc] cost ≤ cost° | R = cost ≤ cost° |
| 7 | = [nil,snoc] R | |

**schedule=(null→nil,pick (schedule×𝟙) snoc), (∀j. 0 ≤ ct(j)); (∀j. 0 ≤ wt(j))** (10.3e)
**⇒ (μX : $\frac{[nil,snag]^\circ}{\ni}$ est(Q) F(X) [nil,snoc]) ⊑ $\frac{bagify^\circ}{\ni}$ est(R)**

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way

the specification — ⦇[nil,snag]⦈ = bagify




| | | |
|---|---|---|
| 1 | $\frac{bagify^\circ}{\ni}$ est(R) | No greedy **reduce** exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole |
| 2 | ⊒ $\frac{[nil,snag]^\circ}{\ni}$ est(Q) F(X) [nil,snoc] | Proposition 10.1: nil and snag have disjoint ranges |
| 3 | ⊒ $\frac{snag^\circ}{\ni}$ est(Q') (X × 𝟙) snoc | pick⊑$\frac{snag^\circ}{\ni}$ est(Q'), a partial function, quadratic in the number of jobs |
| 4 | ⊒ pick (X × 𝟙) snoc | |

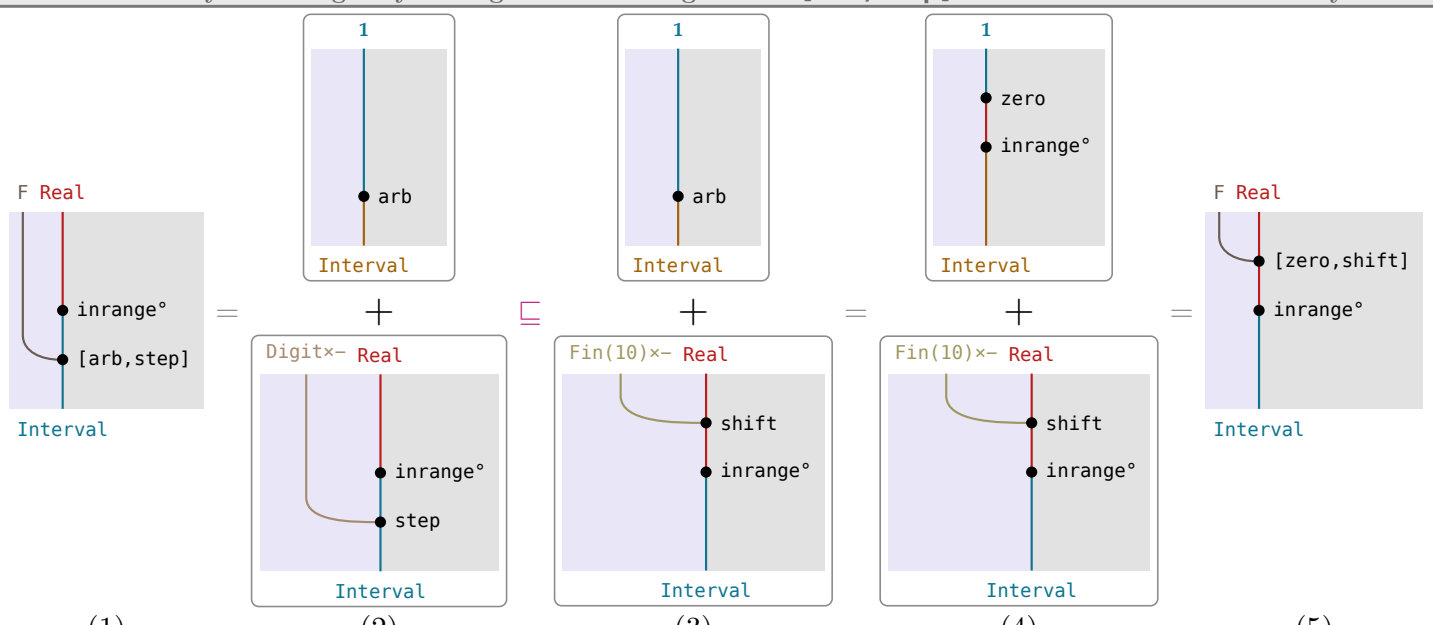
## §10.4. The TeX problem

(10.4a)

| | definition | type | meaning |
|---|---|---|---|
| 1 | `intern ≜ val round` | `[Digit]→[0,2¹⁶)` | the value of a decimal fraction, rounded to `16` binary places |
| 2 | `val ≜ (\|[zero,shift]\|)` | `[Digit]→Real` | the real number a digit list denotes after the decimal point |
| 3 | `shift(d,r) ≜ (d + r)/10` | `Digit×Real→Real` | put digit `d` in front of `r`: `(d+r)/10` |
| 4 | `round° = interval inrange` | `Real→[0,2¹⁶)` | the nearest integer to `2¹⁶r`: `round r=n` iff `2n−1<2¹⁷r<2n+1` |
| 5 | `interval(n) ≜ ((2n−1)/w,(2n + 1)/w)` | `[0,2¹⁶)→ Interval` | the open interval `((2n−1)/2¹⁷, (2n+1)/2¹⁷)` of the reals that round to `n` |
| 6 | `p inrange r ⇔ lo(p) < r ∧ r < hi(p)` | `Interval→Real` | `r` lies strictly inside `(a,b)` |
| 7 | `Legal(p) ≜ 0 < hi(p) ∧ hi(p) < 1 ∧ lo(p) < hi(p)` | `Prop` | (10.9): `0<b<1` and `a<b`; `Interval` is the legal pairs |
| 8 | `x arb p ⇔ lo(p) < 0 ∧ 0 < hi(p)` | `𝟙→Interval` | any legal interval |
| 9 | `step(p) ≜ (shift(π₁(p),lo(π₂(p))),shift(π₁(p),hi(π₂(p))))` | `Digit×Interval→ Interval` | put digit `d` in front of both ends: `((d+a)/10,(d+b)/10)` |
| 10 | `H ≜ (\|[arb,step]\|)°` | `Interval→ [Digit]` | every digit list whose interval the given one is |
| 11 | `xs R ys ⇔ length(xs) ≤ length(ys)` | `[Digit]→[Digit]` | `x` is no longer than `y` |
| 12 | `l ≜ inl` | `𝟙→FX` | the injection of the empty case into `FX=1+(Digit×X)` |
| 13 | `r ≜ inr` | `Digit×X→FX` | the injection of the digit case into `FX=1+(Digit×X)` |
| 14 | `! ≜ ⊸ ()` | `Digit×Interval→ 𝟙` | the one map to `𝟙` |
| 15 | `Q ≜ (l° !° r) ∪ 𝟙` | `F(Interval)→ F(Interval)` | `u` is the empty case and `z` a digit case, or `u=z` |
| 16 | `w ≜ 131072` | `Int` | `2¹⁷` |

**`(|[arb,step]|) ⊑ val inrange°`** (10.4b)

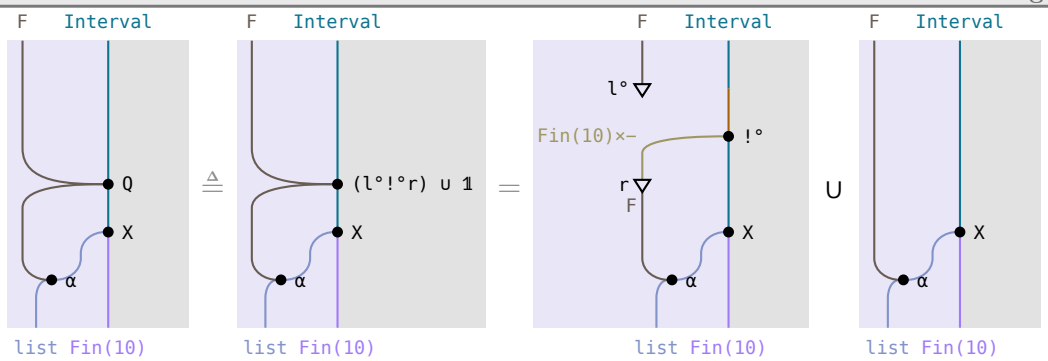
every interval got by folding a decimal's digits with `[arb,step]` has the decimal's value strictly inside it




| | | |
|---|---|---|
| 1 | `F(inrange°) [arb,step]` | `F(S) [arb,step] = [arb,(𝟙 × S) step]` |
| 2 | `= [arb,(𝟙 × inrange°) step]` | `(𝟙 × inrange°) step ⊑ shift inrange°` |
| 3 | `⊑ [arb,shift inrange°]` | `zero inrange° = arb` |
| 4 | `= [zero inrange°,shift inrange°]` | `[zero,shift] Z = [zero Z,shift Z]` |
| 5 | `= [zero,shift] inrange°` | |

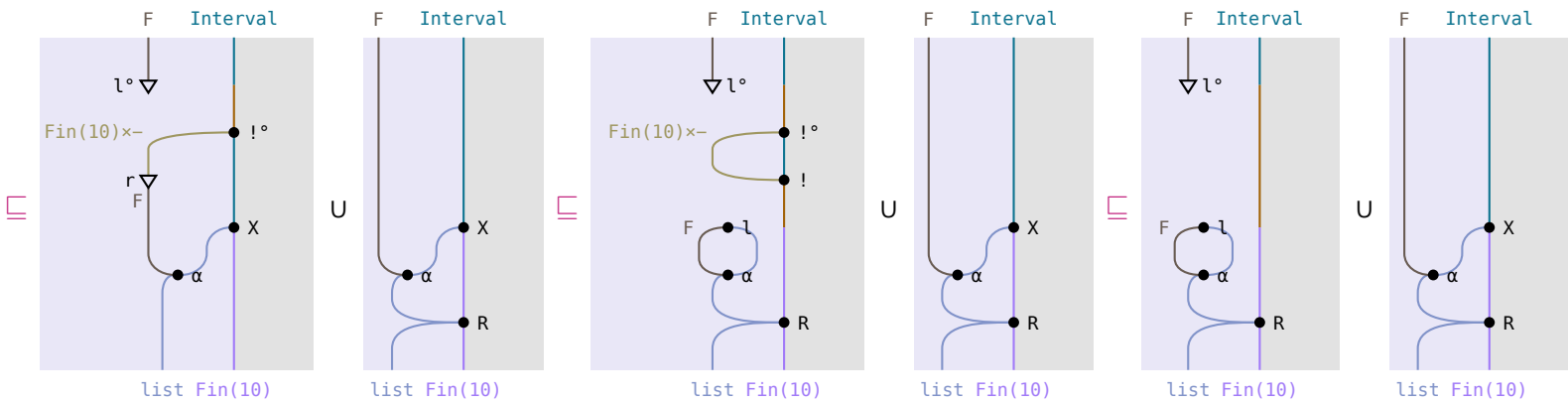
**`Q F(X) α ⊑ F(X) α R`** (10.4c)

choosing by `Q` among the one-step decompositions before building with `F(X)` and `α` yields only decimals that are `R`-related to one built without choosing



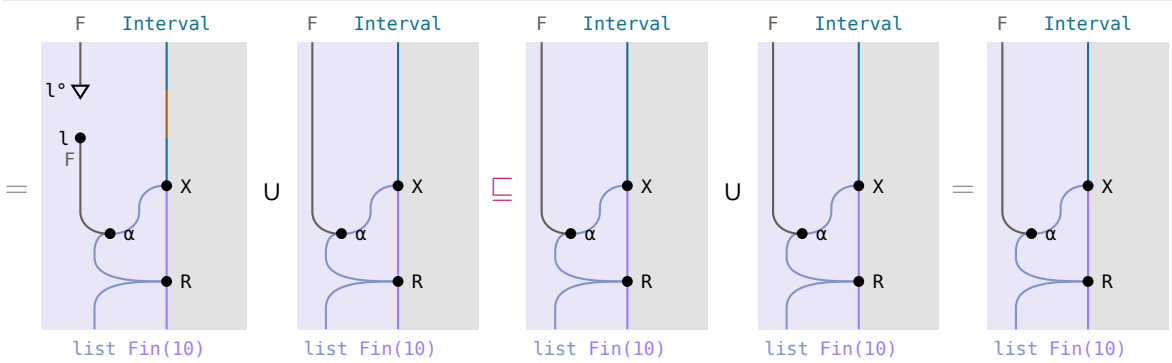

(1) (2) (3)

| | | |
|---|---|---|
| 1 | `Q F(X) α` | `Q ≜ (l° !° r) ∪ 𝟙` |
| 2 | `≜ ((l° !° r) ∪ 𝟙) F(X) α` | `(S ∪ T) R = (S R) ∪ (T R)` |
| 3 | `= (l° !° r F(X) α) ∪ (F(X) α)` | `𝟙 ⊑ R` |




(4) (5) (6)

| | | |
|---|---|---|
| 4 | `⊑ (l° !° r F(X) α) ∪ (F(X) α R)` | `r F(X) α ⊑ ! l α R` |
| 5 | `⊑ (l° !° ! l α R) ∪ (F(X) α R)` | `!° ! ⊑ 𝟙` |
| 6 | `⊑ (l° l α R) ∪ (F(X) α R)` | `l F(X) = l` |

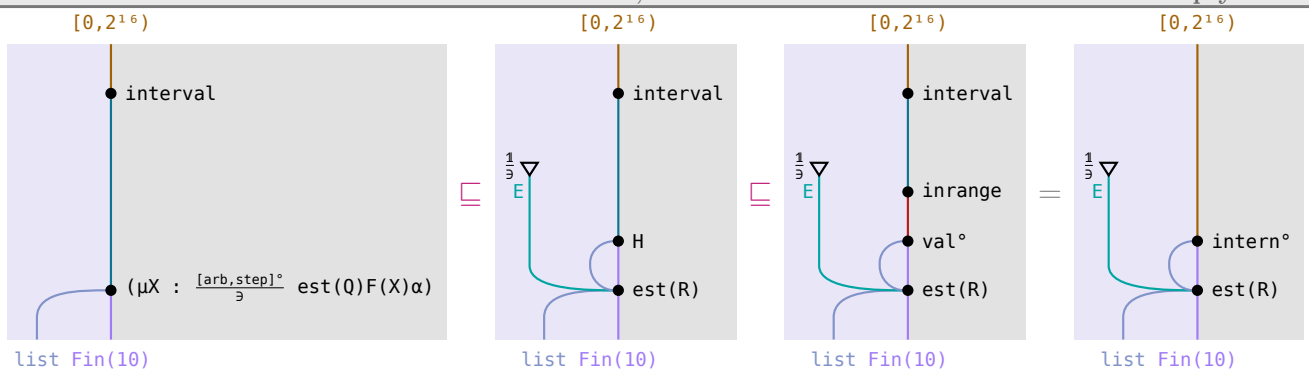



(7) (8) (9)

| | | |
|---|---|---|
| 7 | `= (l° l F(X) α R) ∪ (F(X) α R)` | `l° l ⊑ 𝟙` |
| 8 | `⊑ (F(X) α R) ∪ (F(X) α R)` | `R ∪ R = R` |
| 9 | `= F(X) α R` | |

**extern(n)=f(2n−1,2n+1), interval (μX : $\frac{[arb,step]^\circ}{\ni}$ est(Q) F(X) α) ⊑ $\frac{intern^\circ}{\ni}$ est(R)** (10.4d)

a shortest decimal whose internal representation is the given multiple of $2^{-16}$ is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do




| | | |
|---|---|---|
| 1 | interval (μX : $\frac{[arb,step]^\circ}{\ni}$ est(Q) F(X) α) | $\frac{[arb,step]^\circ}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(R)) α ⊑ $\frac{H}{\ni}$ est(R) |
| 2 | ⊑ interval $\frac{H}{\ni}$ est(R) | $\frac{H}{\ni}$ est(R) ⊑ $\frac{inrange\ val^\circ}{\ni}$ est(R) |
| 3 | ⊑ interval $\frac{inrange\ val^\circ}{\ni}$ est(R) | $\frac{intern^\circ}{\ni}$ = interval $\frac{inrange\ val^\circ}{\ni}$ |
| 4 | = $\frac{intern^\circ}{\ni}$ est(R) | |

(10.4e)

- `f(a,b) = xs ⇔ a < 0 ∧ xs = [] ∨ ¬a < 0 ∧ (∃d. ∃ys. d = ⌊10b⌋ ∧ f(10a−d,10b−d) = ys ∧ xs = [d]⧺ys)`
  `f` gives `(a,b)` the empty decimal when `a<0`, and otherwise the digit `d=⌊10b⌋` followed by what it gives `(10a−d,10b−d)`
- `(μX : [arb,step]° / ∋ est(Q) F(X) α) = f`
  the least solution of the greedy recursion of `interval (μX : [arb,step]° / ∋ est(Q) F(X) α) ⊑ intern° / ∋ est(R)` is `f`
- `d = ⌊10b⌋; e = ⌊10b⌋ ⇒ d = e`
  Exercise 10.12: two digits `d` and `e` with `0<10b−d<1` and `0<10b−e<1` are equal
- `0 < q−p ∧ q < w; p < 0 ⇒ f(p,q) = []`
  the program's first clause: `f(p,q)=[]` when `p<0`
- `0 < q−p ∧ q < w; ¬p < 0 ⇒ f(p,q) = [10q/w]⧺f(10p−w·(10q/w),10q−w·(10q/w))`
  the program's second clause: with `d=(10q) div w`, `f(p,q)` is `d` followed by `f(10p−w·d,10q−w·d)`
- `0 < q−p ∧ q < w; Reach(p,q) ⇒ f(p/w,q/w) = f(p,q)`
  on the pairs the program reaches from `interval n`, the integer `f` at `(p,q)` is a value of the rational `f` at `(p/w,q/w)`
- `interval f = extern`
  `f` after `interval` is the program `extern`, whose value at `n` is `f(2n−1,2n+1)`

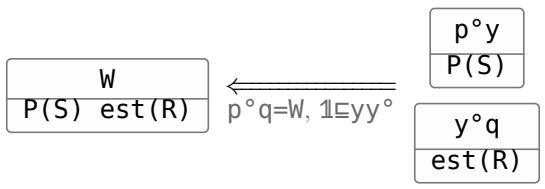
# §A. Appendix

## §A.1. P(S) est(R)=(∋S)∩(∈\(SR°))

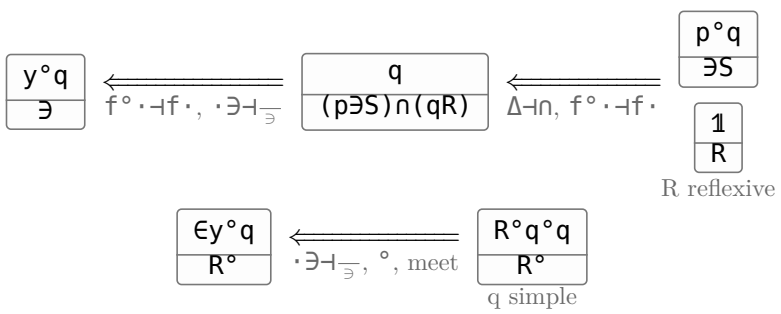
(A.1a)

(p,q) tabulates W≜(∋S)∩(∈\(SR°)), and y≜ $\frac{(p∋S)∩(qR)}{∋}$, a map.

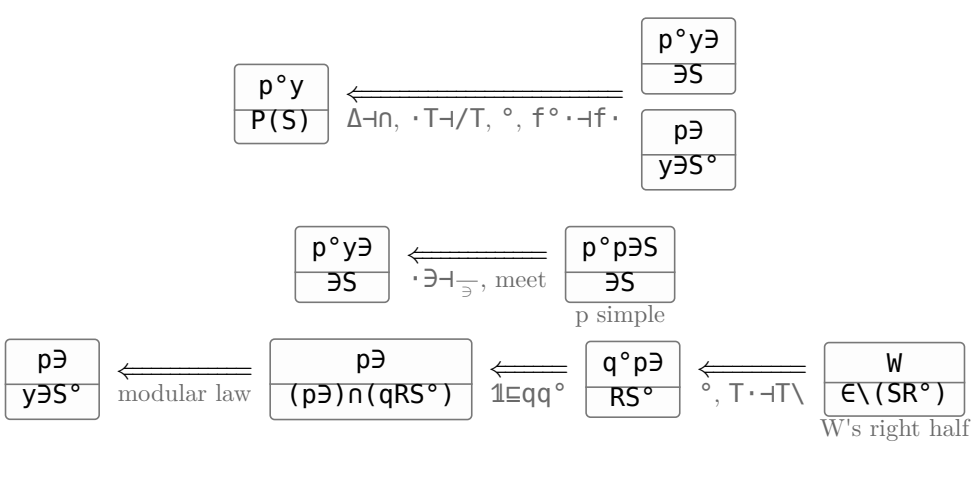
(A.1b)




(A.1c)




(A.1d)



## §A.2. P(est(R)) est(R)=P(dom(est(R))) union est(R)

(A.2a)

P(est(R)) est(R) ═══ dom(est(R)) est(R)=est(R) ═══ P(dom(est(R)) est(R)) est(R)

═══ P a relator ═══ P(dom(est(R)))P(est(R)) est(R)

(7.11) $\overline{\sqsubseteq}$ $R^2 \sqsubseteq R$ ⇒ P(est(R)) est(R) ⊑ union est(R)

P(dom(est(R))) union est(R)

(A.2b)

P(dom(est(R))) union est(R) / ∋est(R) ⟸ 𝟙 ⊑ R ⟸ P(dom(est(R))) union est(R) / P(est(R)) est(R)

⇒ P(S) est(R) = (∋ S) ∩ (∈ \ (S R°)) at S:=est(R), Δ⊣∩, T·⊣T\

∈P(dom(est(R))) union est(R) / est(R)R°

P(dom(est(R))) union est(R) / ∋est(R) ⟸ P(dom(est(R)))⊑𝟙 ═══ union est(R) / ∋est(R)

T(U∩V)⊑TU∩TV, ·∋⊣$\overline{\ni}$, (7.3) union (∈ \ R) = ∈ \ (∈ \ R) ═══ (∋∋)∩(∈\(∈\R°)) / ∋est(R) ⟸ modular law ═══ ∋(∋∩(∈(∈\(∈\R°)))) / ∋(∋∩(∈\R°))

counit of T · ⊣T\

∈P(dom(est(R))) union est(R) / est(R)R° ⟸ ∈ lax natural ═══ dom(est(R))∈union est(R) / est(R)R°

dom(est(R))⊑est(R) est(R)° ═══ est(R) est(R)°∈union est(R) / est(R)R° ⟸ est(R)°⊑∈, ∈∈union⊑∈ ═══ est(R)∈est(R) / est(R)R°

UP of est

## §A.3. Shortest paths on a cylinder, on lists

(A.3a)

| | definition | type | note |
|---|---|---|---|
| 1 | `F(A,X)=A+A×X` | `𝒜→𝒜` | The base functor: one square of the new column, alone or in front of the path so far. |
| 2 | `L=list⁺` | `𝒜→𝒜` | A path is a non-empty list of squares, one per column crossed. |
| 3 | `α` the initial algebra | `Alg(F,μF)` | `α(5)=[5]` and `α(1,[5])=[1,5]`: a path is started by one square, or extended by one. |
| 4 | `N`<br>the `n`-tuple relator | `𝒜→𝒜` | One component per row of a column, so the fold carries `n` answers at once. |
| 5 | `R≜sum≤sum°` | `[ℕ]→[ℕ]` | The cost of a path, which the cheapest minimises. |
| 6 | `setify` | `A[n]→PA` | `setify(1,2,3,4)={1,2,3,4}` — which row a component came from is forgotten. |
| 7 | `moves` | `A[n]→P(A[n])` | `moves(x)={up(x),x,down(x)}` — rotated up, unrotated, rotated down. |
| 8 | `trans` | `P(A[n])→(P A)[n]` | `trans{(a,b,c),(x,y,z)}=({a,x},{b,y},{c,z})` — component `k` of the result is the set of the `k`-th components. |
| 9 | `zip` | `F(a[n])→(F a)[n]` | `zip((1,2,3,4),({[5]},{[6]},{[7]},{[8]}))=((1,{[5]}),(2,{[6]}),(3,{[7]}),(4,{[8]}))`. |
| 10 | `cp≜` $\frac{\texttt{F(1,∋)}}{\texttt{∋}}$ | `F(PA)→PFA` | `cp(1,{[5],[6],[8]})={(1,[5]),(1,[6]),(1,[8])}`. |
| 11 | `gen ≜ F(𝟙,moves trans N(union))`<br>`zip N(cp(F(A,−)) E(α))` | `F(NA,N(P(TA)))→`<br>`N(P(TA))` | `gen((1,2,3,4),({[5]},{[6]},{[7]},{[8]}))` is worked out in `gen ≜ F(𝟙,moves trans N(union))`<br>`zip N(cp(F(A,−)) E(α))` . |
| 12 | `paths ≜ (gen) setify union` | `T(NA)→P(TA)` | `paths[(1,2,3,4),(5,6,7,8)]` is the union of `gen ≜ F(𝟙,moves trans N(union)) zip`<br>`N(cp(F(A,−)) E(α))` 's four sets: 12 paths, 3 from each entry row. |
| 13 | the specification<br>`paths est(R)` | `[ℕ[n]]→[ℕ]` | A cheapest path from the entry side to the exit side. |

### §A.3.1. N(E(L A))

(A.3.1a)

```
1  5
2  6
3  7
4  8
```

```
[(1,2,3,4),(5,6,7,8)] : L N Nat
```

(A.3.1b)

```
gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,−)) E(α))

(gen)[(5,6,7,8)] = ({[5]},{[6]},{[7]},{[8]})

gen((1,2,3,4),({[5]},{[6]},{[7]},{[8]})) =
    ( {[1,5],[1,6],[1,8]},
      {[2,5],[2,6],[2,7]},
      {[3,6],[3,7],[3,8]},
      {[4,5],[4,7],[4,8]} )
```

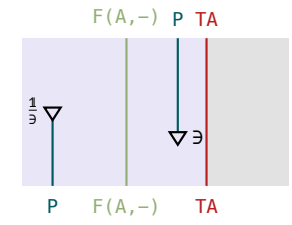
(A.3.1c)

| | | |
|---|---|---|
| 1 | `N` | the 4-tuple, one component per row of the entry column |
| 2 | `E` | the set of paths that can start in that row |
| 3 | `L` | a path is the non-empty list of the squares it crosses |

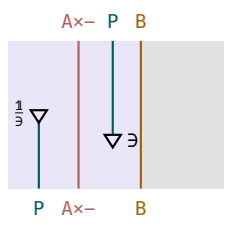
row 1's set has no `[1,7]` — from row 1 only rows 4, 1 and 2 are reachable — and it does have `[1,8]`, because the cylinder glues the bottom row to the top. That is `moves trans N(union)` of `gen` ≜ `F(𝟙,moves trans N(union)) zip N(cp(F(A,−)) E(α))` : component `k` collects the paths of rows `k-1`, `k`, `k+1`.

## §A.3.2. The cross product cp≜$\frac{F(\mathbb{1},\ni)}{\ni}$

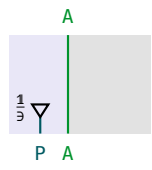
(A.3.2a)




`cp` on all of `F(A,E(L A))`




the `A×−` summand: `∋` picks one path, `𝟙%∋` collects the results




the `A` summand, no `E` to distribute: `𝟙%∋` alone, `a↦{a}`

(A.3.2b)

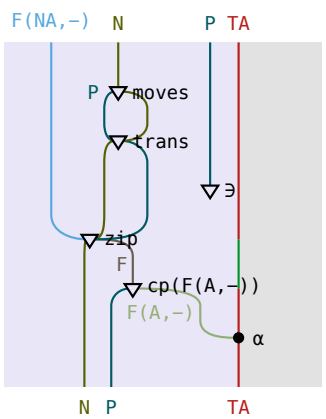
| | type | note |
|---|---|---|
| 1 | `F(A,−)=A+A×−` | one wire carries the whole functor, both summands with it; the `∋` under it is `F(𝟙,∋)` |
| 2 | `∋:E(L A)→L A` | the `∋` inside `F(𝟙,∋)`: a set of paths, one of them — `L A` is one path, `A` one square |
| 3 | `F(𝟙,∋)=𝟙+𝟙×∋` | `:A+A×E(L A)→A+A×L A`, the relator acting on each summand |
| 4 | `∋:E(A+A×L A)→A+A×L A` | the `∋` under the bar: a set of cells, one of them — a different `∋` |
| 5 | `cp:A+A×E(L A)→E(A+A×L A)` | $\frac{R}{\ni}$ turns `R:X→Y` into `X→E Y` |
| 6 | `cp=`$\frac{\mathbb{1}}{\ni}$ `E(F(𝟙,∋))` | what the picture draws: the unit makes the outer `E`, leaving one `∋` — $\frac{R}{\ni}$ = $\frac{\mathbb{1}}{\ni}$ `E(R)` |

(A.3.2c)

```
w      = (1,{[5],[6],[8]})            : A×E(L A)     the right summand
𝟙×∋    : A×E(L A) → A×L A                            𝟙 keeps the square 1, ∋ picks one path
         w ↦ (1,[5]), (1,[6]), (1,[8]) : A×L A        one output per path in w
cp(w)  = {(1,[5]),(1,[6]),(1,[8])}    : E(A+A×L A)   the three of them, collected
```

## §A.3.3. gen=F(𝟙,moves trans N(union)) zip N(cp P(α))

(A.3.3a)



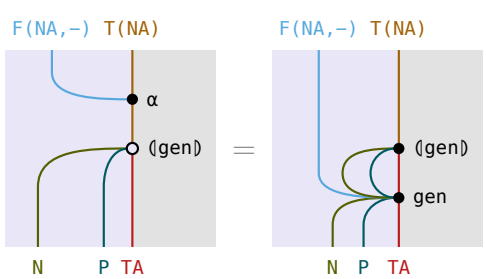
(A.3.3b)

```
gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,−)) E(α))
```

```
u = ((1,2,3,4),({[5]},{[6]},{[7]},{[8]}))      : F(N A,N(E(L A)))
F(𝟙,moves trans N(union))                        𝟙 keeps the column, the path SETS move
  moves({[5]},{[6]},{[7]},{[8]})
   = {({[6]},{[7]},{[8]},{[5]}),
      ({[5]},{[6]},{[7]},{[8]}),
      ({[8]},{[5]},{[6]},{[7]})}                 : E(N(E(L A)))  down, unmoved, up
  trans(that)
   = ({{[6]},{[5]},{[8]}},{{[7]},{[6]},{[5]}},
      {{[8]},{[7]},{[6]}},{{[5]},{[8]},{[7]}}) : N(E(E(L A)))  row k gets rows k-1, k, k+1
  N(union)(that)
   = ({[5],[6],[8]},{[5],[6],[7]},
      {[6],[7],[8]},{[5],[7],[8]})               : N(E(L A))     every path into row k, none dropped
zip(that)
   = ((1,{[5],[6],[8]}),(2,{[5],[6],[7]}),
      (3,{[6],[7],[8]}),(4,{[5],[7],[8]}))       : N(F(A,E(L A)))  each row: its square, and the
                                                                    paths it may be put in front of
N(cp P(α))(that)
   = ({[1,5],[1,6],[1,8]},{[2,5],[2,6],[2,7]},
      {[3,6],[3,7],[3,8]},{[4,5],[4,7],[4,8]}) : N(E(L A))     cp pairs the square with each path,
                                                               α prefixes it: α(1,[5])=[1,5]
```

## §A.3.4. gen is an F-algebra; ⦇gen⦈: α⦇gen⦈=F(𝟙,⦇gen⦈)gen

(A.3.4a)




α puts the column back on the list, then the fold reads all of it — the fold reads the rest under F, then one gen puts the column in front

(A.3.4b)

```
xs = [(1,2,3,4),(5,6,7,8)] = α((1,2,3,4),[(5,6,7,8)])   : L(N A)

⦇gen⦈(xs) = gen((1,2,3,4), ⦇gen⦈[(5,6,7,8)])
```

both halves are in `gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,−)) E(α))` : the tail folds to `({[5]},{[6]},{[7]},{[8]})`, one path per row and each of them one square long, and `gen` on it is `gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,−)) E(α))` 's walk.

## §A.3.5. Q : F(N A,N(L A))→N(L A), an F(N A,−)-algebra: Q=F(𝟙,moves trans N(est(R))) zip N(α)

(A.3.5a)

```
Q : F(N A,N(L A)) → N(L A)                    the fold's algebra: a new column, one path per row

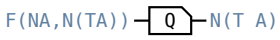
u = ((1,2,3,4),([5],[6],[7],[8]))             : F(N A,N(L A))
F(𝟙,moves trans N(est(R)))                      𝟙 keeps the column, the paths move
  moves([5],[6],[7],[8])
   = {([6],[7],[8],[5]),
      ([5],[6],[7],[8]),
      ([8],[5],[6],[7])}                      : E(N(L A))     down, unmoved, up
  trans(that)
   = ({[6],[5],[8]},{[7],[6],[5]},
      {[8],[7],[6]},{[5],[8],[7]})            : N(E(L A))     row k gets rows k-1, k, k+1
  N(est(R))(that)
   = ([5],[5],[6],[5])                        : N(L A)        the cheapest into each row —
                                                              chosen BEFORE the new square
zip(that)
   = ((1,[5]),(2,[5]),(3,[6]),(4,[5]))        : N(F(A,L A))   each row: its square, and the one
                                                              predecessor that survived
N(α)(that)
   = ([1,5],[2,5],[3,6],[4,5])                : N(L A)        α(1,[5])=[1,5]
```

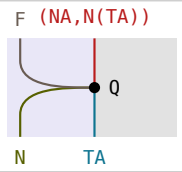
**lax(moves); lax(trans); lax(zip); F(𝟙,moves trans N(est(R))) = [l,pm r]; zip = [N(inl),zip' N(inr)]; inl α = wrap; inr α = cons ⇒ Q = [N(wrap),pm zip' N(cons)]** (A.3.5b)

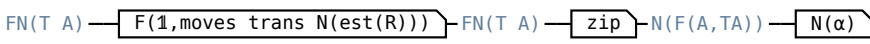
at the last column **Q** starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra **lax(moves); lax(trans); lax(zip); lax(setify); R² ⊑ R; F(𝟙,N(est(R))) Q ⊑ gen N(est(R)) ⇒ ⦇Q⦈ setify est(R) ⊑ paths est(R)** 's last step folds

**circuit** — the fork is F(NA,N(LA))=NA+NA×N(LA)

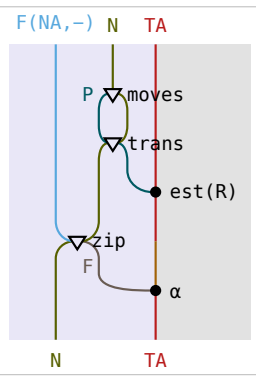
**Hinze–Marsden**

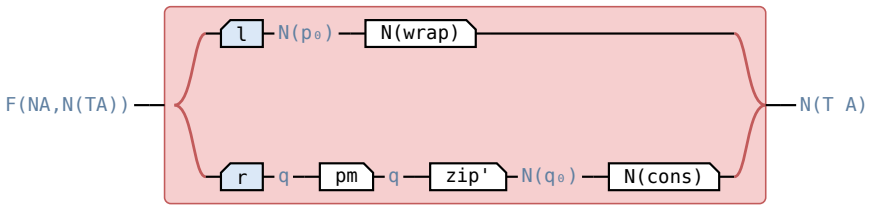



= the fusion condition lax(moves); lax(trans); lax(zip); R² ⊑ R; F(𝟙,est(R)) α ⊑ cp(F(A,−)) E(α) est(R) ⇒ F(𝟙,N(est(R))) Q ⊑ gen N(est(R)) read as a definition,

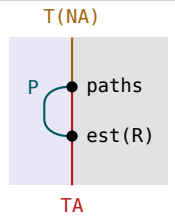
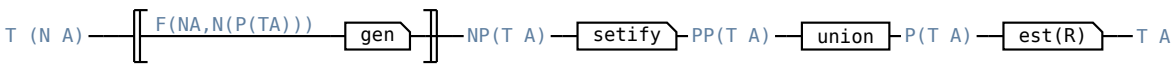
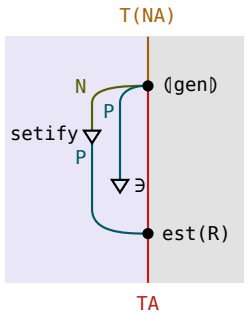
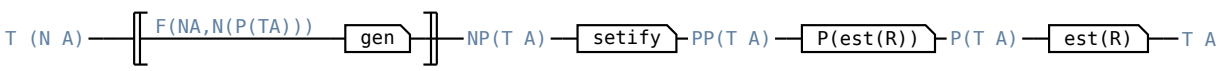
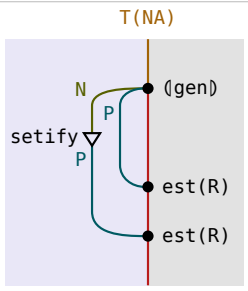
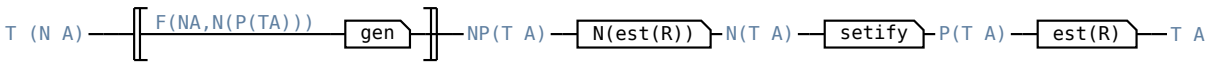
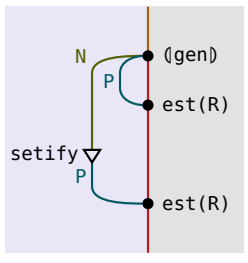
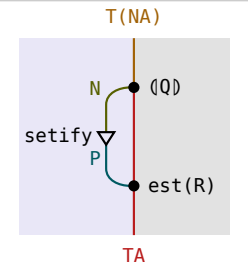



= 


zip=𝟙+zip', α=[τ,cons]

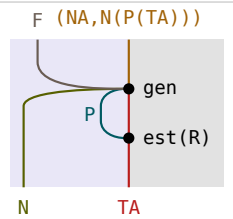
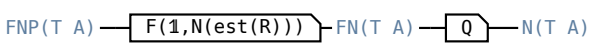
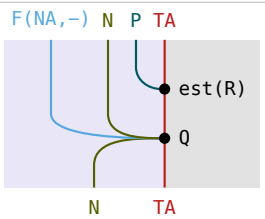
**lax(moves); lax(trans); lax(zip); lax(setify); R² ⊑ R; F(𝟙,N(est(R))) Q ⊑ gen N(est(R))** (A.3.5c)
**⇒ (Q) setify est(R) ⊑ paths est(R)**

shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in `O(n×m)` time; `Q` is `lax(moves); lax(trans); lax(zip); F(𝟙,moves trans N(est(R))) = [l,pm r]; zip = [N(inl),zip' N(inr)]; inl α = wrap; inr α = cons`
⇒ `Q = [N(wrap),pm zip' N(cons)]` 's algebra.

| circuit | Hinze–Marsden |
|---|---|
|  |  |
| =  `paths ≜ (gen) setify union` at `paths` |  |
| ⊒  `P(est(R)) est(R)⊑union est(R)` — a minimum in each set, then a minimum of those; `R` transitive |  |
| ⊒  `setify` lax natural |  |
| ⊒  fusion at `lax(moves); lax(trans); lax(zip); R² ⊑ R; F(𝟙,est(R)) α ⊑ cp(F(A,–)) E(α) est(R)`<br>`⇒ F(𝟙,N(est(R))) Q ⊑ gen N(est(R))` |  |

**`lax(moves); lax(trans); lax(zip); R² ⊑ R; F(𝟙,est(R)) α ⊑ cp(F(A,−)) E(α) est(R)`** (A.3.5d)
**`⇒ F(𝟙,N(est(R))) Q ⊑ gen N(est(R))`**

fusion: the condition for fusion in **`lax(moves); lax(trans); lax(zip); lax(setify); R² ⊑ R; F(𝟙,N(est(R))) Q ⊑ gen N(est(R))`** `⇒ (|Q|) setify est(R) ⊑ paths est(R)` 's last step, used to derive a definition of `Q`.

| circuit | Hinze–Marsden |
|---|---|
|  |  |
| ⊑  <br>(7.13), then `zip`, `trans`, `moves` lax natural |  |

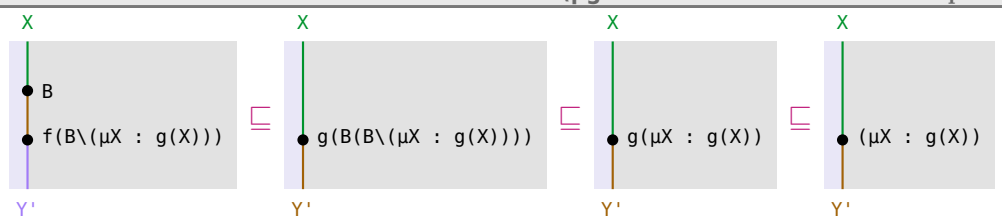
(7.13) is `F(𝟙,est(R))α⊑cp P(α) est(R)`, Theorem 7.1 `Map(f); 𝟙 ⊑ R` `⇒ f° F(R) f ⊑ R ⇔ F(est(R)) f ⊑` $\frac{\mathtt{F(∋)\ f}}{\mathtt{∋}}$ `est(R)` at the map `α` with $\frac{\mathtt{F(𝟙,∋)α}}{\mathtt{∋}}$ `=cp P(α)`: extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.

## §A.4. Fixed-point fusion

**Monotonic(g); (∀Z. B f(Z) ⊑ g(B Z))** (A.4a)

**⇒ B f(B \ (μX : g(X))) ⊑ (μX : g(X))**

if putting B in front of one step of f lands below one step of g with B in front, then B in front of one step of f at **B\μg** lands below the least fixed point of g

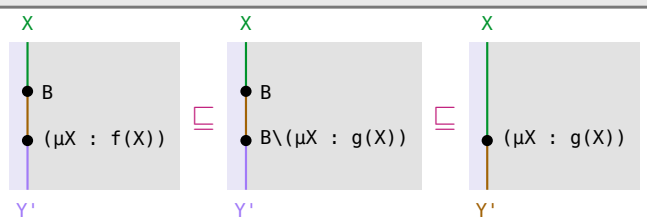



| | | |
|---|---|---|
| 1 | B f(B \ (μX : g(X))) | |
| 2 | ⊑ g(B (B \ (μX : g(X)))) | S (S \ X) ⊑ X |
| 3 | ⊑ g(μX : g(X)) | Monotonic(g)<br>⇒ g(μX : g(X)) ⊑ (μX : g(X)) |
| 4 | ⊑ (μX : g(X)) | |

**Monotonic(g); (∀Z. B f(Z) ⊑ g(B Z))** (A.4b)

**⇒ B (μX : f(X)) ⊑ (μX : g(X))**

if putting B in front of one step of f lands below one step of g with B in front, B in front of the least fixed point of f lands below that of g

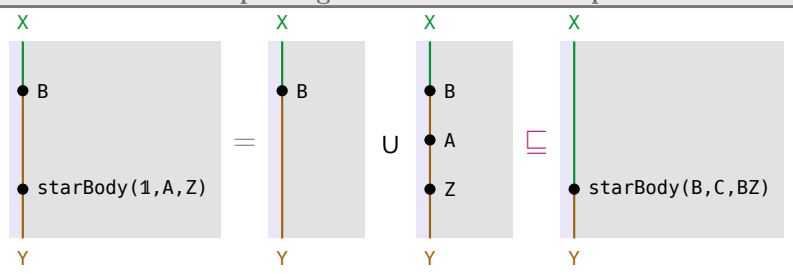



| | | |
|---|---|---|
| 1 | B (μX : f(X)) | Monotonic(g); (∀Z. B f(Z) ⊑ g(B Z))<br>⇒ (μX : f(X)) ⊑ B \ (μX : g(X)) |
| 2 | ⊑ B (B \ (μX : g(X))) | S (S \ X) ⊑ X |
| 3 | ⊑ (μX : g(X)) | |

**B A ⊑ C B** (A.4c)

**⇒ B starBody(𝟙,A,Z) ⊑ starBody(B,C,B Z)**

putting B in front of one step of the A recursion lands below one step of the C·B recursion



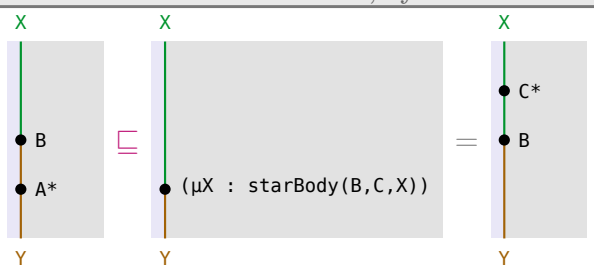

| | | |
|---|---|---|
| 1 | B starBody(𝟙,A,Z) | B (S ∪ T) = (B S) ∪ (B T) |
| 2 | = B ∪ (B A Z) | B A ⊑ C B |
| 3 | ⊑ starBody(B,C,B Z) | |

**B A ⊑ C B** (A.4d)

**⇒ B A* ⊑ C* B**

the same **B A* ⊑ C* B** as **B A ⊑ C B**

**⇒ B A* ⊑ C* B** , by fusion: **A*** and **C* B** are least fixed points of **starBody(𝟙,A)** and **starBody(B,C)**




(1) (2) (3)

| | | |
|---|---|---|
| 1 | B A* | B A ⊑ C B<br>⇒ B A* ⊑ (μX : starBody(B,C,X)) |
| 2 | ⊑ (μX : starBody(B,C,X)) | C* B = (μX : starBody(B,C,X)) |
| 3 | = C* B | |